

Unless stated otherwise, all problems are worth 3 points and answers should be stated as exact fractions or decimals rounded to three decimal places. You must show your work to receive credit.

1. A card is dealt from a well-shuffled, standard deck of 52 cards. Find the probability of being dealt each of the following (write your answer as a fraction).

- a. an ace

$$p(A) = 4/52 = 1/13$$

- c. an ace or a spade

$$p(A \cup \spadesuit) = 4+13-1/52 = 16/52 = 4/13$$

- b. the ace of spades

$$p(A \cap \spadesuit) = 1/52$$

- d. A four of spades

$$p(4 \cap \spadesuit) = 4/52 = 1/13$$

2. A coins is flipped and a six-sided die is tossed. Find the following:

- a. the sample space

$$S = \{1H, 1T, 2H, 2T, 3H, 3T, 4H, 4T, 5H, 5T, 6H, 6T\}$$

- b. the probability of getting a head AND rolling a three.

$$p(H \cap 3) = 1/12$$

- c. [2 points] the odds of getting a head AND rolling a three.

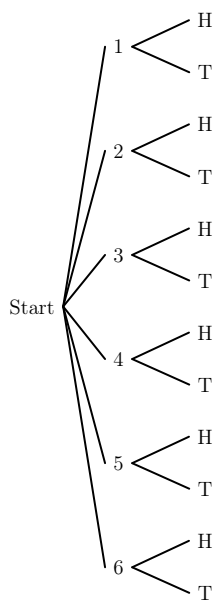
$$o(H \cap 3) = 1 : 11$$

- d. the probability of getting a head OR rolling a three.

$$p(H \cup 3) = 2+6-1/12 = 7/12$$

- e. [2 points] the odds of getting a head OR rolling a three.

$$o(H \cup 3) = 7 : 5$$



3. [6 points each] Complete the following:

- a. The MHU librarian has found that the number of books checked out by students is as follows:

Number of Books	0	1	2	3
proportion	25%	50%	15%	10%

Find the expected number of books checked out by a student.

$$\mathbb{E}V = 0 \cdot .25 + 1 \cdot .50 + 2 \cdot .15 + 3 \cdot .10 = 1.1$$

- b. You have been asked to play a game with a six-sided die. The rules are as follows.

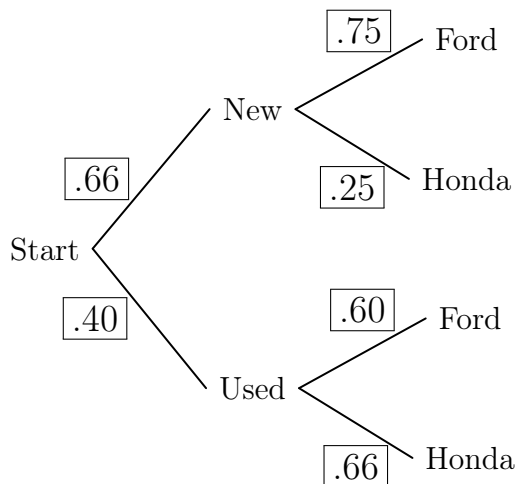
- If you roll a 1 or 2 you win \$50.
- If you roll a 3 you lose \$10.
- If you roll a 4, 5, or 6 you lose \$20

Find the expected value of the game.

$$2/6 \cdot 50 + 1/6 \cdot (-10) + 3/6 \cdot (-20) = 5$$

4. Approximately 66% of the cars at Big Bill's Cars are new and 34% are used. Of the new cars, 75% are Fords and the rest are Hondas. While 40% of the used cars are Fords and the rest are Hondas.

- a. Complete the tree diagram describing the probability



- b. find the probability that a new car is a Ford.

$$p(N \mid F) = .75$$

- c. find the probability that a used car is a Ford.

$$p(U \cap F) = .40 \cdot .60 = .24$$

- d. find the probability that a car on the lot is a Ford.

$$\begin{aligned} p((U \cap F) \cup (N \cap F)) \\ = .40 \cdot .60 + .66 \cdot .75 = .735 \end{aligned}$$

5. Below are the results of a Gallup poll in which workers were asked “Would you prefer a male or female boss?”

	Prefer Male boss	Prefer Female boss	No preference
Male	29%	18%	51%
Female	40%	27%	32%

- a. What is the probability that a worker prefers a female boss?

$$p(PF) = \frac{.18 + .27}{2.0} = .225$$

- b. What is the probability that a worker prefers a male boss given that they are female?

$$p(PM | F) = .40$$

- c. What is the probability that someone is female given that they prefer a male boss?

$$p(F | PM) = \frac{.40}{.29 + .40} \approx .58$$

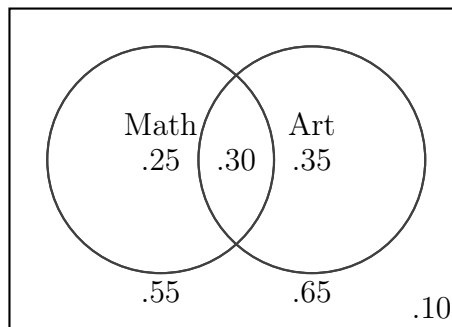
6. Of the students who are Biology majors, 55% are taking Math, 65% are taking Art, and 30% are taking both courses. Find the probability of the following:

- a. a biology major is not taking Math

$$1 - .55 = .45$$

- b. a biology major is taking Math or Art

$$.55 + .65 - .30 = .90$$



- c. a biology major is not taking Math and not taking Art.

$$1 - .90 = .10$$

7. A die is rolled. The probability of rolling a 5 is $1/6$, and the probability of rolling a 5 given that a 4 or higher is rolled is $1/3$.
The probability of rolling a 4 or 5 is $2/6 = 1/3$, and the probability of rolling a 4 or 5 given that an even number is rolled is $1/3$. Justify your answers.

- a. Are the events “rolling a 5” and “rolling a 4 or higher” independent or dependent?

Dependent: $p(5) = 1/6 \neq 1/3 = p(5 \mid 4 \cup 5 \cup 6)$

- b. Are the events “rolling a 5” and “rolling a 4 or higher” mutually exclusive or non-mutually exclusive?

Non-mutually exclusive: if you roll a 5 you also roll a 4 or higher.

- c. Are the events “rolling a 5” and “rolling an even number” mutually exclusive or non-mutually exclusive?

Mutually exclusive: $5 \notin \{2, 4, 6\}$

- d. Are the events “rolling a 4 or 5” and “rolling an even number” independent or dependent?

Independent: $p(4 \cup 5) = 1/3 = p(4 \cup 5 \mid 2 \cup 4 \cup 6)$

8. [6 points each] A jar contains 30 jellybeans. 12 are red and 18 are blue. Jane randomly picks 5 jellybeans. Find the probability of each of the following. **You only need to write the formulas.**

- a. all 5 are blue.

$$\frac{{}_{18}C_5}{{}_{30}C_5}$$

- c. 2 red and 3 blue.

$$\frac{{}_{12}C_2 \cdot {}_{18}C_3}{{}_{30}C_5}$$

- b. exactly 4 are blue.

$$\frac{{}_{18}C_4 \times {}_{12}C_1}{{}_{30}C_5}$$

- d. at least 4 are blue.

$$\frac{{}_{18}C_4 \times {}_{12}C_1 + {}_{18}C_5 \times {}_{12}C_0}{{}_{30}C_5}$$